

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

In triangle XYZ , angle Z is a right angle and the length of $\overline{YZ}$ is **24** units. If $\tan X = \frac{12}{35}$, what is the perimeter, in units, of triangle XYZ ?

Answer

- A. 188
- B. 168
- C. 84
- D. 71

Correct Answer: B

Rationale

Choice B is correct. It's given that angle Z in triangle XYZ is a right angle. Thus, side YZ is the leg opposite angle X and side XZ is the leg adjacent to angle X . The tangent of an acute angle in a right triangle is the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. It follows that $\tan X = \frac{YZ}{XZ}$. It's given that $\tan X = \frac{12}{35}$ and the length of side YZ is **24** units. Substituting $\frac{12}{35}$ for $\tan X$ and **24** for YZ in the equation $\tan X = \frac{YZ}{XZ}$ yields $\frac{12}{35} = \frac{24}{XZ}$. Multiplying both sides of this equation by $35(XZ)$ yields $12(XZ) = 24(35)$, or $12(XZ) = 840$. Dividing both sides of this equation by **12** yields $XZ = 70$. The length XY can be calculated using the Pythagorean theorem, which states that if a right triangle has legs with lengths of a and b and a hypotenuse with length c , then $a^2 + b^2 = c^2$. Substituting **70** for a and **24** for b in this equation yields $70^2 + 24^2 = c^2$, or $5,476 = c^2$. Taking the square root of both sides of this equation yields $\pm 74 = c$. Since the length of the hypotenuse must be positive, $74 = c$. Therefore, the length of XY is **74** units. The perimeter of a triangle is the sum of the lengths of all sides. Thus, $(74 + 70 + 24)$ units, or **168** units, is the perimeter of triangle XYZ .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This would be the perimeter, in units, for a right triangle where the length of side YZ is **12** units, not **24** units.

Choice D is incorrect and may result from conceptual or calculation errors.